

Review Article

Dynamic GIF generation in WebGIS applications: addressing connectivity constraints in offshore and remote environments

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ABSTRACT

This paper examines the practical case for dynamically generating graphics interchange format (GIF) outputs within web geographic information system (WebGIS) platforms as a response to a concrete operational problem: the difficulty of delivering large, complex geospatial datasets — particularly NetCDF and JSON files — to users operating in bandwidth-constrained or offline environments such as deep ocean voyages and remote maritime deployments. The proposed approach involves server-side rendering of time-series geospatial data into animated GIF sequences that users can download in advance and consult without any live network connection. Key findings indicate that this method significantly extends the practical reach of WebGIS platforms, improves operational safety for navigators and researchers at sea, and does so at substantially lower bandwidth cost than raw data transfer. The paper further establishes the technical, compatibility, and intellectual property considerations that govern effective GIF deployment.

Keywords: WebGIS, Animated GIF, NetCDF, Geospatial visualization, Offline data access, Bandwidth-constrained environments

INTRODUCTION

WebGIS platforms have transformed how geospatial and oceanographic data are disseminated to scientific and operational communities. These platforms typically store and serve data in formats such as network common data form (NetCDF) or JavaScript object notation (JSON) — formats optimized for computational fidelity, multi-variable storage, and programmatic access.¹

However, these same formats present a fundamental obstacle when users require access to spatial data under conditions of poor or absent internet connectivity.

THE PROBLEM: DATA DELIVERY TO CONNECTIVITY-CONSTRAINED USERS

The problem is most acute in deep ocean areas. Vessels operating in offshore and open-ocean regions rely on

satellite communication links that are characterized by low data throughput, high latency, and considerable per-megabyte cost. Downloading raw NetCDF files — which can reach hundreds of megabytes for multi-day, multi-variable forecasts — is often economically or technically infeasible over such connections. Interactive WebGIS interfaces fare no better in this context, as they require a live connection to dynamically render layers and respond to user queries in real time. The result is that researchers, navigators, and operational personnel in offshore environments are frequently cut off from the most current geospatial intelligence precisely when it is most needed.

Beyond connectivity costs, the technical burden of client-side data handling compounds the problem. Working directly with NetCDF or large JSON payloads requires specialized software libraries — such as xarray, GDAL, or dedicated oceanographic toolkits — that are not universally available on the range of devices deployed in

maritime or field settings. Ruggedized laptops, tablets, and integrated bridge systems may support only standard operating system utilities, making the rendering of raw scientific data formats impractical without additional software installation. This represents a significant gap between data availability at the server level and genuine data accessibility at the point of use.

TECHNICAL APPROACH: DYNAMIC GIF GENERATION FROM WEBGIS DATA

The solution adopted in this system is the server-side, on-demand generation of animated GIF files directly from the underlying NetCDF and JSON datasets held within the WebGIS platform. Rather than requiring users to download raw data and render it locally, the platform performs all data parsing, spatial rendering, and frame composition on the server, producing a self-contained animated GIF that encapsulates the temporal evolution of the selected geophysical variable over the forecast or analysis period.

Frame composition and temporal encoding

The GIF89a specification — introduced in 1989 — provides native support for encoding multiple image frames within a single binary file, along with timing metadata that governs the display interval between frames.² This capability is directly exploited in the GIF generation pipeline. For each time step in the source dataset, the platform renders the relevant spatial variable — for example, wind speed and direction, wave height, sea surface temperature, or current velocity — onto a geo-registered map canvas. The resulting frames are then assembled in chronological sequence and encoded with appropriate inter-frame timing to produce a smooth animation that accurately represents the temporal progression of the phenomenon.

The rendered output for wind speed and direction demonstrates the operational context: a forecast issued on 21 May 2026 covering South Asia's coastal and offshore zones, with a color-coded overlay indicating wind intensity and an animated playback interface permitting users to step through time at adjustable frame rates.

This interactive rendering is the basis from which the downloadable GIF is generated — the same sequence of rendered map frames, captured and packaged into a portable animated file.

Compression and color palette management

GIF files employ Lempel-Ziv-Welch (LZW) lossless compression, a scheme that achieves particularly efficient file sizes for images with limited, discrete color palettes.³ The format's restriction to a maximum of 256 indexed colors per frame is well-matched to the visual characteristics of standard WebGIS outputs, which typically represent continuous geophysical variables using

discrete color ramps or banded palettes — such as a wind speed scale ranging from blue (low) through green and yellow to red (high).

By designing the platform's color rendering to work within the indexed-color constraint, the system avoids the banding artifacts that can otherwise degrade the visual quality of GIF outputs while maintaining the compact file sizes that are essential for bandwidth-limited download.

In practice, the resulting GIF file sizes are substantially smaller than equivalent sequences of individual PNG frames or compressed video files, and far smaller than the raw NetCDF or JSON datasets from which they are derived. This compression efficiency is the primary technical enabler of practical data delivery over satellite links and other low-bandwidth channels.

Download integration within the WebGIS interface

The GIF generation functionality is exposed to users directly within the platform's web interface via a dedicated download control — the 'download GIF' button in the upper right of the application. This placement reflects a deliberate design decision to make pre-departure GIF download a first-class, easily discoverable feature rather than an ancillary export option.

A parallel 'Download Video' control provides access to MPEG-format equivalents for users in contexts where larger file sizes are acceptable; however, the GIF option is specifically positioned for use cases where file compactness and universal compatibility are paramount. The animated GIF format's self-contained character — all frames and timing encoded in a single binary — means that once downloaded, the file requires no additional software, codecs, or runtime dependencies to render correctly on any device.

IMPROVED OFFLINE USABILITY FOR REMOTE USERS

The most operationally significant outcome of the GIF generation approach is the substantial improvement in data accessibility for users in offline or near-offline environments. A downloaded GIF constitutes a fully self-contained analytical artifact: once it resides on a user's device, it functions entirely independently of any network connection, server infrastructure, or software environment beyond a standard image viewer.

For navigation officers and shipboard scientists departing on extended voyages, this represents a qualitative shift in capability. Prior to departure — while still within range of adequate connectivity — a navigator may download a 72-hour animated forecast of swell height and current velocity for the planned transit route. Once at sea, this file can be consulted at any time without requiring a satellite data session. The animated sequence, looping continuously, conveys the temporal evolution of ocean conditions in a

perceptually intuitive form that a static chart or tabular data extract cannot replicate. This continuous-loop behavior — a characteristic native to the GIF format with no analogous mechanism in standard video players — enhances the salience of cyclic and periodic patterns such as tidal rhythms, diurnal wind cycles, and recurring circulation features.

Shipboard scientists benefit similarly. A researcher conducting sampling operations in a region of dynamic sea surface temperature gradients may pre-download animated temperature or chlorophyll animations before leaving port, then reference them to guide station selection and contextualize in situ measurements during the voyage. This capability is not contingent on the vessel's satellite link being active or affordable at the moment of need — the visualization is already on the device, ready for immediate consultation.

The platform's compatibility profile further ensures that these benefits are not restricted by the device or operating system in use. GIF rendering is incorporated natively into every major web browser, desktop operating system, and mobile platform without the need for additional codecs or plugins.⁴ An oceanographer consulting a downloaded GIF on a ruggedized bridge terminal, a personal laptop, or a tablet encounters no technical barrier to viewing the animation — a practical advantage that formats requiring specialized rendering software cannot offer.

MAIN RESULTS AND OPERATIONAL IMPLICATIONS

The integration of dynamic GIF generation into the WebGIS platform produced several concrete results that validate the approach as a solution to the offline data access problem.

First, the file size reduction achieved through LZW-compressed GIF encoding rendered pre-departure download of multi-day forecast animations practically feasible over satellite links that would be wholly inadequate for raw NetCDF or JSON data transfer. The economic and technical barriers that previously prevented offshore users from carrying current forecast data on their devices were substantively lowered.

Second, the animated format preserved the temporal information content of the underlying dataset in a form that is directly perceptible to human users without any analytical processing. Researchers and operators do not need to understand the structure of NetCDF files or write code to extract time slices — they observe the phenomenon evolving visually, in the spatial context of the map, at the pace set by the animation timing.

This democratization of access — extending sophisticated spatiotemporal data visualization to users without geoscientific software expertise — represents a significant broadening of the platform's effective user base.

Third, the self-contained, software-independent nature of the downloaded GIF enabled its embedding into reports, briefing documents, and presentation materials in a way that preserves the animated, temporal quality of the visualization. Researchers producing voyage reports or operational summaries could incorporate directly relevant animated visualizations without requiring recipients to have access to the originating WebGIS platform.

Fourth, the intellectual property framework governing GIF outputs generated from open geospatial datasets — including those published under national meteorological agency open data policies and international scientific data sharing agreements — was found to be straightforward: outputs algorithmically produced from open-access source data may generally be treated as original works and distributed accordingly, subject to source attribution requirements.⁵ This clarity reduces the legal uncertainty that might otherwise inhibit data sharing in operational and scientific contexts.

CONCLUSION

The dynamic generation and download of animated GIFs within WebGIS platforms addresses a real and significant gap in geospatial data accessibility: the inability of users in bandwidth-constrained or offline environments to receive and utilize the rich time-series spatial data that modern oceanographic and meteorological platforms hold. By moving the computational work of data rendering to the server — leveraging the GIF format's native animation support, LZW compression efficiency, and universal device compatibility — the system converts complex, multi-gigabyte datasets into compact, portable, self-contained visualizations that can be carried onto vessels, into the field, or to any location beyond the reach of a reliable internet connection.

The approach does not require changes to the underlying data infrastructure, does not impose new software requirements on end-users, and is compatible with open-access data licensing frameworks. For WebGIS platforms serving operational maritime, oceanographic, and environmental science communities, it represents a practically motivated, technically sound, and immediately deployable strategy for extending platform utility to the full range of users and environments these systems are intended to serve.

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